Runge-Kutta method

The Runge-Kutta method is a self-starting numerical method which is comely use in solving ODE.

There are many types of this method depending on the order of the drifter.

1. Second-order R-K method is

$$y_{n+1} = y_n + \frac{1}{2}(k_1 + k_1)$$

Where $k_1 = hf(x_n, y_n)$

$$k_2 = hf(x_n + h, y_n + k_1)$$

2. Third order R-K method is

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 4k_2 + k_3)$$

Where $k_1 = hf(x_n, y_n)$

$$k_2 = hf\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}k_1\right)$$

$$k_3 = hf(x_n + h, y_n - k_1 + 2k_2)$$

3. Forth order R-K

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Where $k_1 = hf(x_n, y_n)$

$$k_2 = hf\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}k_1\right)$$

$$k_3 = hf\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}k_2\right)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

Which is generally used in solving ODE numerically.

Ex (1):- solve the following differential eq. using the 2nd order R-K method

$\frac{dy}{dx} = \frac{1}{2}y, y(0) = 1, h = 1$

Solu.

$$y_1 = y_0 + \frac{1}{2}(k_1 + k_2)$$

$$k_1 = hf(x_0, y_0) = 1f(0,1) = 1\left(\frac{1}{2} \cdot 1\right) = 0.5$$

$$k_2 = hf(x_0 + h, y_0 + k_1) = 1f(1,1.5) = 1 \cdot \frac{1.5}{2} = 0.75$$

$$\therefore y_1 = 1 + \frac{1}{2}(0.5 + 0.75) = 1 + \frac{1.25}{2} = 1 + 0.625 = 1.625$$

Ex (2):- by R-K fourth order solve the following eq. $y' = x^2 - 2xy$, when $y(1) = 0, h = 0.2, n = 2$

Solu.

$$f(x, y) = y' = x^2 - 2xy$$

$$k_1 = hf(x_0, y_0) = 0.2\left(1^2 - 2(1)(0)\right) = 0.2$$

$$k_2 = hf(x_0 + 0.1, y_0 + 0.1) = 0.2f(1.1,0.1) = 0.2\left((1.1)^2 - 2(1.1)(0.1)\right)$$
$$= 0.198$$

$$k_3 = 0.2f(1.1,0.099) = 0.2\left((1.1)^2 - 2(1.1)(0.099)\right) = 0.2(1.21 - 0.2178)$$
$$= 0.1984$$

$$k_4 = 0.2f(1.2,0.1984) = 0.2\left((1.2)^2 - 2(1.2)(0.1984)\right) = 0.2(1.44 - 0.4762)$$
$$= 0.1928$$

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2(k_2 + k_3) + k_4)$$

$$= 0 + \frac{1}{6}(0.2 + 2(0.198 + 0.1984) + 0.1928) = 0 + 0.1976 = 0.1976$$

$$x_1 = x_0 + h = 1 + 0.2 = 1.2$$

$$y_2 = y_1 + \frac{1}{6}(k_1 + 2(k_2 + k_3) + k_4)$$

$$k_1 = hf(x_1, y_1) = 0.2f(1.2,0.1976) = 0.2\left((1.2)^2 - 2(1.2)(0.1976)\right) = 0.1932$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = 0.2f\left(1.3,0.1976 + \frac{0.1932}{2}\right)$$

$$= 0.2\left((1.3)^2 - 2(1.3)(0.2942)\right)$$

$$= 0.2(1.69 - 0.76492) = 0.1850$$

$$k_3 = 0.2f(1.3,0.2901) = 0.2\left((1.3)^2 - 2(1.3)(0.2901)\right) = 0.2(1.69 - 0.7543)$$

$$= 0.1871$$

$$k_4 = 0.2f(1.4,0.3847) = 0.2\left((1.4)^2 - 2(1.4)(0.3847)\right) = 0.2(1.96 - 1.0772)$$

$$= 0.1766$$

$$y_2 = 0.1976 + \frac{1}{6}(0.1932 + 2(0.185 + 0.1871) + 0.1766) = 0.1976 + 0.1857$$

$$= 0.3833$$